

Portfolio Optimization: Analyzing the Impact of Investment Constraints Using Monte Carlo Simulation

Scenario 2: Diversified 4-Asset Portfolio (NVDA, XOM, TLT, GLD)

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1. Executive Summary and Problem Definition

This analysis investigates the critical role of investment constraints specifically, the allowance of **short selling** on the potential risk and return profiles achievable in a diversified portfolio. We utilize a **Monte Carlo simulation** approach to map the entire feasible investment space for a four-asset portfolio, allowing for a clear, intuitive comparison between a traditional **long-only** strategy and a more complex **long-plus-short** strategy.

The portfolio is intentionally structured to provide diversification across distinct asset classes:

Ticker	Asset Name	Asset Class	Rationale
NVDA	NVIDIA Corporation	Technology Equity	High-growth, high-volatility exposure.
XOM	Exxon Mobil Corporation	Energy Equity	Value-oriented, cyclical equity exposure.
TLT	iShares 20+ Year Treasury Bond ETF	Bonds	Fixed-income, often counter-cyclical, low-volatility anchor.
GLD	SPDR Gold Shares	Gold / Commodity	Inflation hedge and store-of-value commodity exposure.

The core objective is to visually and quantitatively demonstrate how relaxing the **no-shorting constraint** fundamentally alters the boundaries and shape of the **feasible set** of portfolios in the risk–return plane.

2. Methodology: Monte Carlo Simulation and Portfolio Metrics

Investment Constraints

To explore the entire investment opportunity set, thousands of random portfolio weight combinations were generated under two distinct sets of constraints:

1. **Long-Only Portfolios:**

- Weights are non-negative
- The sum of all weights equals one
- *Simulation Technique:* Weights were efficiently sampled using a **Dirichlet distribution**.

2. **Shorts Permitted Portfolios (Long + Short):**

- Weights can be both positive (long) or negative (short selling allowed).
- The sum of all weights equals one.
- *Constraint:* A **soft leverage filter** was applied to the simulated weights to exclude extremely unrealistic or highly levered portfolios, ensuring practical relevance.

Portfolio Calculation

For every simulated portfolio, the key metrics of expected return and volatility were calculated using the foundational principles of **Mean–Variance Portfolio Theory**

3. Model Parameters and Assumptions

The simulation relies on the following forward-looking assumptions regarding the expected performance and co-movement of the four assets. These parameters define the theoretical shape of the efficient frontier.

Asset-Specific Parameters

Asset	Expected Return (μ)	Volatility (σ)
NVDA	22.0%	50.0%
XOM	8.0%	25.0%

TLT	6.0%	15.0%
GLD	5.0%	16.0%

Correlation Matrix

The assumed correlation structure is critical, as it drives the benefits of diversification:

This matrix highlights:

- A moderate positive correlation between the two equity components (**NVDA** and **XOM**).
- Desirable mild negative correlations between the equities and the **TLT** bond ETF, which is the principal source of **diversification benefit**.
- Weak and varied correlations involving **GLD**, suggesting its independent role as a potential risk reducer.

```
# Correlation matrix (stylized diversification)
corr = np.array([
    [ 1.00,  0.40, -0.20,  0.10], # NVDA
    [ 0.40,  1.00, -0.10,  0.05], # XOM
    [-0.20, -0.10,  1.00,  0.10], # TLT
    [ 0.10,  0.05,  0.10,  1.00], # GLD
], dtype=float)
```

4. Analysis of Results

The key outcome is a visual representation of the simulated portfolios in the risk–return space, contrasting the two investment regimes.

4.1. Long-Only Portfolios

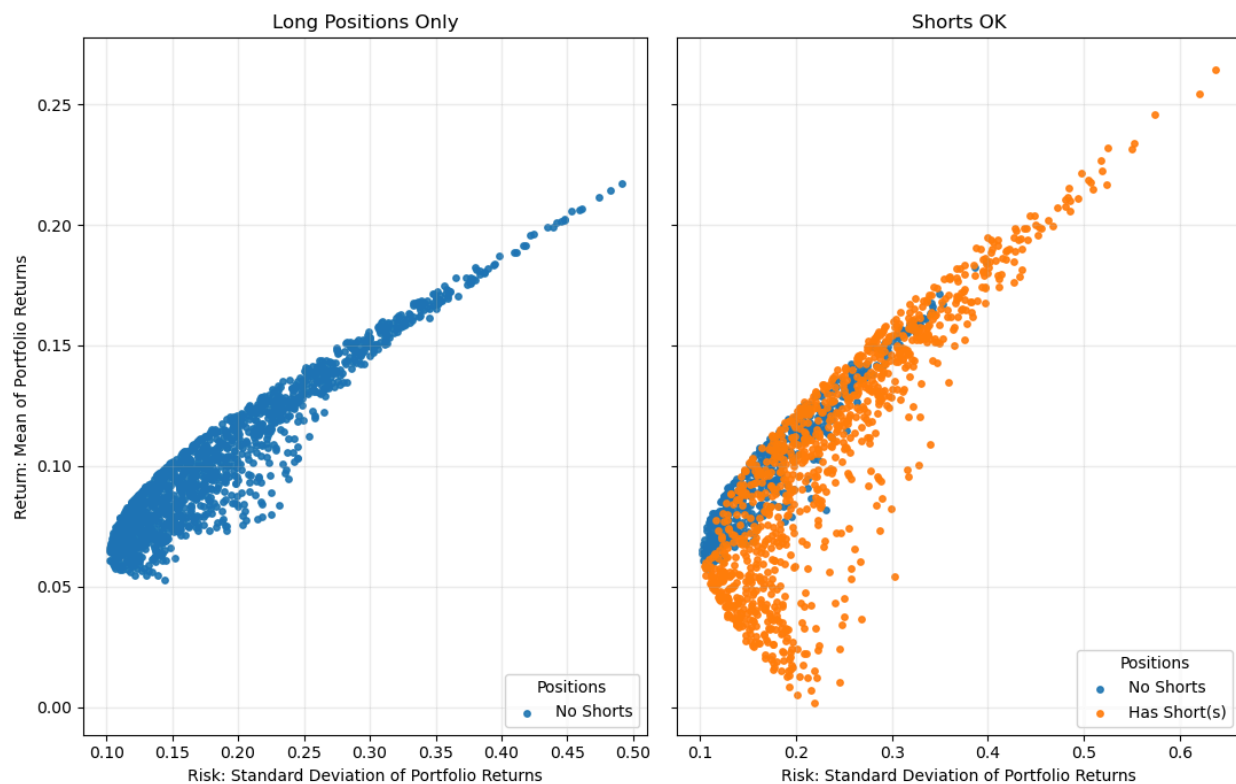
The set of portfolios generated under the long-only constraint forms a relatively **compact and contained** opportunity set .

- **Risk Profile:** The overall risk (volatility) is naturally bounded, as no leverage is introduced through short selling.
- **Return Profile:** Expected returns are capped by the highest returning asset, and the maximum return is constrained by the necessity of holding a diversified, un-levered portfolio.
- **Efficient Frontier:** The upper-left boundary of this compact set defines the classic **efficient frontier**, representing the optimal trade-off between risk and return for a conservative investor.

4.2. Shorts Permitted Portfolios

When short selling is allowed, the feasible investment space **expands dramatically** in both dimensions .

- **Expanded Set:** The simulated orange points (representing portfolios with short positions) extend the opportunity set significantly, particularly toward the **higher-risk and higher-return** quadrants.
- **Leverage and Hedging:** Short selling allows the portfolio manager to effectively **short low-performing assets** to fund higher-performing ones (increasing expected return) or to take leveraged directional bets (increasing risk).
- **Theoretical Alignment:** This expansion confirms the principles of **Mean–Variance Theory**, where removing the non-negativity constraint on weights fundamentally broadens the mathematically optimal set of portfolios.



5. Discussion and Key Takeaways

This experiment effectively illustrates the trade-offs inherent in portfolio construction based on investment policy.

The Role of Investment Constraints

The contrast between the two investment regimes is stark:

- **Long-Only** portfolios are inherently **more conservative and risk-managed**. They are appropriate for investors with lower risk tolerance or those operating under strict regulatory mandates that prohibit margin or short selling.
- **Long-Plus-Short** portfolios offer access to a **broader, potentially more profitable efficient frontier**. However, this comes with the necessity of accepting **significantly higher volatility** and the added operational complexity of short selling, margin calls, and regulatory oversight.

The Benefit of Diversification

The shape of the efficient frontier in both scenarios confirms the value of **asset class diversification**. By combining assets with negative or low positive correlations (equities, bonds, and gold), the resulting efficient frontier is positioned **further to the left** (lower risk) than if only highly correlated assets were used.

Conclusion

The **Monte Carlo simulation** proves to be a powerful and intuitive tool for visualizing the complex landscape of portfolio possibilities. It allows investors to clearly map the risk–return trade-offs associated with different asset allocations and the critical decision of whether to employ leverage via short selling. For a professional investor, the choice between these two strategies is a fundamental decision that balances the pursuit of maximal returns against the practical realities of risk capacity and operational management.